

Cross-Lingual Hallucination Drift in LLMs: Does It Depend on Task Type?

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Abstract

Large Language Models (LLMs) are known to hallucinate in non-English languages, but whether this *cross-lingual hallucination drift* is task-dependent remains an open question. We evaluate **Aya Expansive 8B** across English, Spanish, Italian, and Swahili on two structurally different tasks: factual QA (TruthfulQA) and commonsense reasoning (XCOPA), using GPT-4o-mini as an LLM-as-a-Judge evaluator. Italian is included in both benchmarks as a confound-control language, enabling a within-language test that isolates task type from language identity.

Our main finding is that drift is strongly task-dependent. TruthfulQA drift is negligible for Spanish ($\Delta\text{HR} = -2.66$ pp, $p = 0.60$) and Italian ($\Delta\text{HR} = +1.34$ pp, $p = 0.80$). XCOPA Italian drift is similarly small ($+1.33$ pp, $p = 0.68$), while XCOPA Swahili drift is catastrophic ($+90.67$ pp, $p < 0.001$). The pairwise drift interaction score $\Phi = -93.33$ pp (Spanish/TruthfulQA vs. Swahili/XCOPA) is highly significant ($\chi^2 = 45.44$, $p < 0.001$). A within-Italian test (fixing language and varying only task) confirms task-dependency without any language-identity confound ($\chi^2 = 16.98$, $p < 0.001$).

Qualitative error analysis shows that Swahili XCOPA hallucinations are predominantly *incoherent* (27% of failures, avg. 185 tokens), unlike other cells which produce short wrong answers. Dual-judge validation (GPT-4o-mini vs. Claude claude-sonnet-4-5 vs. human annotators on 50 Italian samples) yields $\kappa = 0.662$ for TruthfulQA/Italian but near-chance agreement ($\kappa = 0.065$) for XCOPA/Italian, revealing inherent difficulty in judging commonsense outputs. We also release an interactive Streamlit dashboard for exploring all results.

1 Introduction

Hallucination, defined as generating content that is factually incorrect or unfaithful, is a well-

documented failure mode of large language models (Alansari et al., 2025). As LLMs are deployed globally, a natural concern is whether hallucination rates are systematically higher in non-English languages, particularly low-resource ones with sparse training data (Blasi et al., 2022). This phenomenon is called *cross-lingual hallucination drift*: the change in hallucination rate when a model operates in a non-English language.

A key recent study by ul Islam et al. (2025) measured token-level hallucination across 30 languages and 6 LLM families, finding that length-normalized hallucination rates are *uncorrelated* with language resource level. This surprising result, however, was established on a single task type: open-domain, knowledge-intensive long-form QA. Whether it holds for structurally different tasks is unexplored.

Factual recall (TruthfulQA) and commonsense reasoning (XCOPA) make different demands on the model. TruthfulQA requires retrieving stored parametric knowledge; XCOPA requires selecting a culturally plausible cause or effect from world knowledge anchored in the target language. These tasks may expose fundamentally different multilingual failure modes.

We formally define the **Drift Interaction Score**:

$$\Phi_l = \Delta\text{HR}_{l, \text{TruthfulQA}} - \Delta\text{HR}_{l, \text{XCOPA}}$$

where $\Delta\text{HR}_{l,t} = \text{HR}_{l,t} - \text{HR}_{\text{en},t}$. A large $|\Phi_l|$ indicates task-dependent drift for language l .

Contributions.

- We run a six-cell evaluation (en/es/it on TruthfulQA; en/it/sw on XCOPA) where Italian serves as a shared language enabling a confound-controlled within-language test.
- We confirm task-dependent drift via both a cross-task test ($\chi^2 = 45.44$, $p < 0.001$) and a clean within-Italian test ($\chi^2 = 16.98$, $p < 0.001$).

- We provide qualitative error analysis, an XCOPA causal-direction breakdown, and dual-judge reliability validation with manual inspection.
- We release an interactive Streamlit dashboard for result exploration.

2 Related Work

Cross-lingual hallucination. [ul Islam et al. \(2025\)](#) is our most direct point of comparison. They find no correlation between language resource level and hallucination in long-form QA, using LaBSE-based semantic similarity as the evaluation signal. Our work tests whether this null result generalizes across task types. [Guerreiro et al. \(2023\)](#) study hallucination in neural machine translation, observing incoherence patterns in low-resource language pairs that foreshadow our Swahili XCOPA findings.

Multilingual benchmarks. TruthfulQA ([Lin et al., 2022](#)) provides adversarial factual questions designed to elicit false beliefs held by humans. [Qin et al. \(2024\)](#) survey multilingual LLM capabilities and identify commonsense reasoning as a task type particularly sensitive to language-specific cultural grounding.

LLM-as-a-Judge. Evaluating generation with a strong LLM judge is now standard practice for open-ended tasks ([Kang et al., 2024](#)). We follow this methodology and add a dual-judge reliability check with human annotation to validate our primary GPT-4o-mini judge.

Hallucination evaluation. [Li et al. \(2023\)](#) introduce HaluEval, a hallucination benchmark for knowledge-grounded tasks, noting that hallucination types vary substantially across task categories, consistent with our error analysis finding that incoherence dominates in Swahili XCOPA while wrong-answer errors dominate in TruthfulQA.

3 Methods

3.1 Datasets

TruthfulQA ([Lin et al., 2022](#)). 817 questions across 38 domains designed to elicit confident false beliefs. We use the English original and Spanish and Italian translations from the `alexandrainst/m_truthfulqa` HuggingFace dataset.

XCOPA ([Ponti et al., 2020](#)). A cross-lingual commonsense causal reasoning benchmark spanning 11 languages. Each example presents a premise and two candidate causes or effects; the model selects the more plausible one. We use the English (SuperGLUE COPA), Italian, and Swahili splits. Spanish is not available in XCOPA.

Italian as confound control. A key methodological choice is to include Italian in *both* benchmarks. This enables a within-language test: fixing language to Italian and varying task, any significant difference in hallucination pattern must be attributable to task type, not language identity or resource level. This directly addresses the confound noted by our midway reviewer.

Sampling. We sample 150 examples per cell with random seed 42, yielding 900 total evaluation instances across six cells.

3.2 Target Model

We evaluate **Aya Expanse 8B** (CohereLabs), a multilingual instruction-tuned model explicitly trained for high-resource and low-resource languages. We run 4-bit NF4 quantization (BitsAndBytes, double quant, `float16` compute dtype) with greedy decoding and a maximum of 200 new tokens. Prompts are zero-shot and written in the target language: TruthfulQA uses a free-form answer format and XCOPA uses a forced-choice (A/B) format. Inference runs on a Google Colab T4 GPU (15.8 GB VRAM), with an estimated total runtime of approximately 1.5–2 hours for all six cells.

3.3 LLM-as-a-Judge

Primary judge. **GPT-4o-mini** receives each (question, model response) pair and labels it *Hallucinated* or *Faithful* with a brief natural-language justification ([Kang et al., 2024](#)). GPT-4o-mini was chosen for its strong multilingual comprehension across all four languages. Any response receiving an ERROR label is automatically retried; all 900 items received valid labels after retry.

Dual judge and manual validation. To validate judge reliability on Italian, a language the team can read directly, we run a second judge pass using **Claude claude-sonnet-4-5** on a stratified 50-sample subset per Italian cell. The same 50 samples are then manually inspected to produce ground-truth human labels, following the reviewer’s suggestion to combine LLM dual-judging with manual

inspection (Kang et al., 2024). This yields three-way inter-rater agreement (GPT-4o-mini, Claude, human).

3.4 Evaluation Metrics

We compute the **hallucination rate**:

$$\text{HR}_{l,t} = \frac{\# \text{ Hallucinated}}{N_{l,t}} \times 100 (\%)$$

where $N_{l,t} = 150$ for all cells. Drift from English is $\Delta\text{HR}_{l,t} = \text{HR}_{l,t} - \text{HR}_{\text{en},t}$. Cohen’s κ is used for inter-rater agreement. All code and outputs are publicly available on GitHub.¹

3.5 Statistical Tests

- **Mann-Whitney U**: tests whether per-item hallucination rates differ between English and each non-English language within a task.
- **Chi-squared** (2×2): (a) aggregate cross-task test on all non-English responses pooled; (b) within-Italian test on Italian-only responses.
- **Fisher’s exact**: robustness check on the same tables.

3.6 Error Categorization

We categorize GPT-4o-mini judge justifications using keyword heuristics on the reason text into five types: *incoherent* (rambling, off-topic, or structurally garbled), *code-switching* (mid-response language switch), *fabricated* (hallucinated entities or facts), *wrong answer* (plausible but incorrect), and *other*.

3.7 Streamlit Dashboard

We built an interactive dashboard using Streamlit for result exploration, accessible by running `streamlit run app.py`. The dashboard has five pages: (1) a KPI summary with Φ scores and the full six-cell HR table; (2) interactive charts including HR bars, drift comparison, and a heatmap; (3) a “New Analyses” page with tabs for XCOPA cause/effect breakdown, dual-judge κ visualization, and error category charts; (4) an example browser for browsing individual model responses filtered by task, language, and label; and (5) a reason analysis page with hallucination category distributions.

¹<https://github.com/bhoomi02-ai/Cross-Lingual-Hallucination-Drift-in-LLMs>

Task	Lang	HR (%)	ΔHR	Avg Tok
TruthfulQA	en	27.33	—	81.7
TruthfulQA	es	24.67	−2.66 pp	108.0
TruthfulQA	it	28.67	+1.34 pp	93.8
XCOPA	en	8.00	—	89.2
XCOPA	it	9.33	+1.33 pp	117.4
XCOPA	sw	98.67	+90.67 pp	185.0

Table 1: Hallucination rate, drift from English, and average response token count. $N = 150$ per cell. Avg Tok = average tokens per response (all labels).

4 Results

4.1 Hallucination Rates and Drift

Table 1 shows hallucination rates across all six cells. TruthfulQA rates are similar across English (27.33%), Spanish (24.67%), and Italian (28.67%), with small and statistically non-significant drifts. XCOPA shows the same near-English behavior for Italian (9.33%, +1.33 pp) but catastrophic collapse for Swahili (98.67%, +90.67 pp).

4.2 Task-Dependent Drift

Table 2 presents all statistical tests.

Cross-task aggregate. Pooling all non-English responses, the hallucination outcome is not independent of task type ($\chi^2 = 45.44$, $p < 0.001$). The pairwise drift interaction score $\Phi = -93.33$ pp (Spanish/TruthfulQA vs. Swahili/XCOPA) far exceeds our pre-specified minimum effect threshold of $|\Phi| \geq 2$ pp.

Within-Italian confound test. Restricting to Italian only, task type strongly predicts hallucination outcome ($\chi^2 = 16.98$, $p < 0.001$, Fisher OR=0.26, $p < 0.001$). TruthfulQA/Italian (HR = 28.67%) and XCOPA/Italian (HR = 9.33%) have very different absolute rates even though both drift from English by only ≈ 1.3 pp. This test eliminates the language-identity confound entirely: the same language produces starkly different hallucination profiles across task types.

4.3 Figures

Figure 1 shows hallucination rates across all six cells. Figure 2 shows ΔHR per non-English cell, making the task-type interaction visually clear: Spanish and Italian bars are near zero for both tasks, while Swahili XCOPA towers above all others.

Test	Comparison	Stat.	p	Sig.
Mann-Whitney U	TruthfulQA: en vs es	11550.0	0.600	No
Mann-Whitney U	TruthfulQA: en vs it	11100.0	0.798	No
Mann-Whitney U	XCOPA: en vs it	11100.0	0.683	No
Mann-Whitney U	XCOPA: en vs sw	1050.0	<0.001	Yes
Chi-squared (all non-EN)	Aggregate ($\Phi = -46.67$ pp)	45.44	<0.001	Yes
Fisher exact (robustness)	Aggregate	3.23	<0.001	Yes
Chi-squared lang = it	it/TruthfulQA vs it/XCOPA	16.98	<0.001	Yes
Fisher exact lang = it	it/TruthfulQA vs it/XCOPA	0.26	<0.001	Yes

Table 2: Statistical test results. The bottom block restricts to Italian only, controlling for language identity.

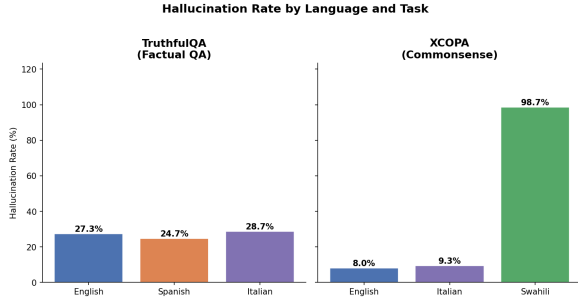


Figure 1: Hallucination rates by language and task. Italian behaves comparably to English in both tasks. Swahili XCOPA is near-total (98.67%).

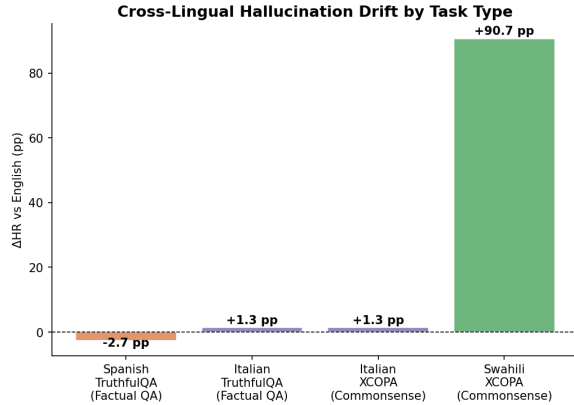


Figure 2: Cross-lingual drift (Δ HR vs. English) per cell. Italian shows near-zero drift in both tasks; Swahili XCOPA shows catastrophic drift (+90.67 pp).

4.4 Token Verbosity

When hallucinating on XCOPA in Swahili, Aya generates an average of 185 tokens per response, approximately twice the English XCOPA average (89 tokens) and well above any TruthfulQA cell (82–108 tokens). This pattern is absent for TruthfulQA in any language, where hallucinated and faithful token counts are comparable. It suggests that Swahili XCOPA failures are not confident short wrong answers but *incoherent rambling*, corroborated by the error analysis below.

Lang	Type	N	HR (%)	p
en	cause	72	11.11	—
en	effect	78	5.13	
it	cause	73	9.59	1.00
it	effect	77	9.09	
sw	cause	56	96.43	0.27
sw	effect	94	100.00	

Table 3: XCOPA hallucination rates by question type. p = Fisher’s exact test within language. No significant cause/effect difference in any language.

4.5 XCOPA Cause vs. Effect

Table 3 breaks XCOPA hallucination by question type. For English and Italian, cause and effect rates differ by ≤ 1 pp and neither difference is significant ($p = 1.0$ and $p = 0.27$ respectively). Swahili shows near-total hallucination in both directions (cause: 96.43%; effect: 100.0%), confirming that the Swahili collapse is not localized to one causal direction but reflects a global failure on this language–task pairing.

4.6 Error Analysis

Table 4 shows hallucination categories per cell. Across TruthfulQA and XCOPA English and Italian, the dominant error type is *wrong answer* (the model gives a plausible but incorrect response). Swahili XCOPA is a clear outlier: 40 of 148 hallucinations (27%) are *incoherent*, consistent with the token verbosity finding. No cell shows meaningful fabrication or code-switching.

4.7 Dual-Judge Agreement

Table 5 shows inter-rater agreement on 50 Italian samples per cell. For TruthfulQA/Italian, agreement is moderate: GPT-4o-mini vs. Claude $\kappa = 0.662$, GPT-4o-mini vs. human $\kappa = 0.597$, Claude vs. human $\kappa = 0.712$. For XCOPA/Italian, all pairwise κ values are near chance (GPT vs. Claude

Task	Lang	Fab.	Incoh.	W.Ans	Total
TruthfulQA	en	0	0	14	41
TruthfulQA	es	1	1	11	37
TruthfulQA	it	1	0	15	43
XCOPA	en	0	0	5	12
XCOPA	it	0	0	6	14
XCOPA	sw	0	40	5	148

Table 4: Hallucination error categories. Fab. = fabricated; Incoh. = incoherent; W.Ans = wrong answer. *Other* omitted for space. N = total hallucinated items per cell.

Cell	n	κ G/C	κ G/H	κ C/H
TruthfulQA/it	50	0.662	0.597	0.712
XCOPA/it	50	0.065	0.256	0.065

Table 5: Cohen’s κ for GPT-4o-mini (G), Claude claude-sonnet-4-5 (C), and human (H) on 50 Italian samples per cell. Raw agreement: TruthfulQA/it = 0.84; XCOPA/it = 0.38.

$\kappa = 0.065$, Claude vs. human $\kappa = 0.065$, GPT vs. human $\kappa = 0.256$).

The low κ on XCOPA/Italian is interpretively important. Since XCOPA/Italian has only 14 hallucinated responses out of 150 (9.33%), most labels are *Faithful* and boundary cases are sparse. Two strong LLM judges genuinely disagree on these edge cases, suggesting that judging commonsense forced-choice outputs is harder than judging factual QA. This does not threaten the main results: the high-hallucination Swahili cell that drives our primary finding produces unambiguously incoherent output.

5 Discussion

Task type dominates language resource level.

Our results provide strong and consistent evidence that hallucination drift is task-dependent. TruthfulQA shows near-zero drift for Spanish and Italian (both mid-resource languages with substantial training data coverage), and we expect the same would hold for other mid-resource languages. XCOPA, however, shows catastrophic drift for Swahili even though Italian (also non-English) behaves comparably to English. The decisive variable is task type, not language resource level.

Why does XCOPA Swahili collapse? Commonsense causal reasoning in XCOPA requires selecting a culturally plausible cause or effect from world knowledge anchored in the target language. When

the model cannot produce a coherent answer in Swahili, it cannot even fall back on a short wrong answer; it produces verbose, incoherent output instead. This contrasts with TruthfulQA hallucinations, which are typically confident but wrong short answers. The failure mode suggests the model’s Swahili XCOPA competence lies below the threshold needed for coherent forced-choice output, not merely below the threshold for accuracy.

The Italian confound test is the cleanest result.

The within-Italian chi-squared ($\chi^2 = 16.98$, $p < 0.001$) shows that task type predicts hallucination outcome even when language is held constant. Notably, $\Phi_{it} = +0.01$ pp, meaning Italian drifts by approximately the same amount from English in both tasks ($\approx +1.3$ pp). The significance of the within-Italian test does not come from differential drift but from the fact that the two tasks have very different absolute hallucination rates (28.67% vs. 9.33% for Italian). This is the purest demonstration that task structure, not language, determines the magnitude of the hallucination problem.

Implications for prior work.

ul Islam et al. (2025) found no correlation between language resource level and hallucination in long-form QA. Our TruthfulQA results agree. But our XCOPA Swahili result challenges the generalization of that finding: task type is a major confound. Studies measuring hallucination drift on a single task type risk drawing language-level conclusions that are actually task-specific. Future work should evaluate drift across multiple task types before attributing cross-lingual differences to language properties.

Limitations.

First, we evaluate a single model; results may differ for models with different multilingual training coverage. Second, Spanish and Swahili do not co-occur in the same benchmark, so the cross-task Φ comparison conflates language and task to some degree. The within-Italian test mitigates but does not fully resolve this, as Italian is a mid-resource language; a truly low-resource language appearing in both benchmarks would be ideal. Third, our error categorization uses keyword heuristics on English judge reasons, which may be less reliable for non-English error descriptions.

6 Conclusion

We measured cross-lingual hallucination drift in Aya Expanse 8B across six language–task cells covering factual QA (TruthfulQA) and commonsense

reasoning (XCOPA) in English, Spanish, Italian, and Swahili. Hallucination drift is strongly task-dependent: factual QA shows negligible drift for all languages tested, while commonsense reasoning shows catastrophic collapse for Swahili. A within-Italian test (same language, both tasks) confirms this is a property of task type. Error analysis shows Swahili XCOPA failures are incoherent rather than plausibly wrong, and dual-judge validation reveals that commonsense forced-choice outputs are harder to judge reliably than factual QA responses. Our findings suggest that task type should be treated as a first-class variable in future cross-lingual hallucination studies.

Author Contributions

Bhoomika Monthy Rajashekar (Data & Infrastructure) handled dataset download and sampling for all six language–task cells including Italian, set up the HuggingFace pipeline and prompt templates, and managed the GitHub repository and data branch.

Chun Hsu (Evaluation & Metrics) designed the LLM-as-a-Judge prompt, implemented the hallucination rate, ΔHR , Φ , and per-language Φ_l metrics for the within-Italian confound test, and built the Streamlit interactive dashboard.

Devinn Chi (Experiments) ran zero-shot inference with Aya Expanse 8B across all six cells with 4-bit NF4 quantization, managed output logging and the failed-response retry pipeline, and ran GPT-4o-mini judge calls including Italian labels.

Anagha P Krishna (Analysis & Writing) conducted all statistical significance tests, performed dual-judge validation with Claude and manual annotation of Italian samples, carried out error categorization and the XCOPA cause vs. effect breakdown, and wrote the final report.

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